BLÜ DENNIS

Junior Developer | Kraaifontein, Cape Town, South Africa
bludennis17@gmail.com | +27 64 766 9310 | LinkedIn | GitHub | workwithme.org.za

SUMMARY

I didn't take the usual path into software development. No formal campus, no fixed syllabus, just a decision to learn from the best teachers I could find. From 2023 to 2025 I studied online through Harvard, MIT, Georgia Tech and others, choosing courses from beginner to advanced level and finishing each one with 80% and above. Along the way my thinking changed from being comfortable with consistency towards becoming comfortable with growth, solving problems in ways I hadn't considered before. The result is a stack of projects: AI experiments, programming applications, databases, security work and websites which grew out of that shift. They reflect steady progress and a foundation I'm prepared to build on in a professional setting.

SKILLS

Programming Languages: C#, Python, Java, JavaScript, TypeScript, PHP, PowerShell, Bash, HTML, CSS, JSON/XML

Frameworks & Tools: Git (branching), ASP.NET, .NET 8,Core Entity Framework, Visual Studio, REST API, Design API Integration, Django, GitHub, VS Code, WordPress, Elementor, jQuery, WooCommerce

Databases: SQL Server, T-SQL, MSSQL, PostgreSQL, MySQL, SQLite

Cloud & DevOps: Microsoft Azure, AWS (EC2, Lambda, S3, IAM, CloudFormation), Azure DevOps, CI/CD concepts, Azure Services

Security Concepts: IAM, PAM, Zero Trust, backup & disaster recovery, Solid Security, cryptography

Key Areas: Front-end development, back-end development, code review, application debugging, API integration, HTTP Standards

EDUCATION

National Senior Certificate
Stellenberg High School — Cape Town, South Africa | 2022

EMPLOYMENT HISTORY

Full-Stack Developer *Temo Digital, Cape Town, South Africa*
June 2026 – Present

- Building Eduman, a React-based school information management system covering learners, staff, finance, assessments, attendance, communication, and incident management across 30+ modules, tracked against Azure DevOps work items.
- Shipped standalone modules including Incident Management, Parent Management, Subject Attendance, Educator Module, Communication (eThetha), Textbook Management, and Finance, each delivered as a working page against a tracked work item.
- Built shared UI infrastructure (dock navigation, glass-surface components, quick actions, listing modals) reused across 30+ pages to keep the app visually and behaviourally consistent.
- Designed and built a theming system for Orchard Core CMS in C#/.NET as a bonus feature on a larger BizSolutions client project, shipping 6 distinct themes with a shared library of 40+ widgets, each restyled per theme so a site's widgets adapt automatically to whichever theme the user has equipped.
- Implemented the widget architecture and theme-switching logic against Orchard Core's content and rendering pipeline.

Website Developer (Independent Contractor) *Superb Digital, Cape Town, South Africa*
April 2025 – February 2026

- Designed and developed WordPress websites for 5 small businesses using Elementor, PHP, HTML, and CSS. Clients went from zero online enquiries to an average of 10–15 inbound leads per month within 4–6 weeks of launch.
- Maintained 15 live client websites with weekly updates, security checks, and bug fixes. Delivered all work within a consistent one-hour service window with zero missed deadlines.

- Implemented Solid Security protocols across all client deployments and reduced reported security incidents to zero.
- Performed code reviews on custom PHP and CSS implementations to enforce consistent development standards across the team.
- Configured domains and SSL certificates via cPanel for every new site launch, ensuring 100% went live securely and development-ready from day one.

PROJECTS

Programs *with Harvard University*

A selection from 13 projects spanning from searching algorithms to neural networks. Full project set available on GitHub.

Traffic Sign Classifier

Tools: Python · TensorFlow/Keras · OpenCV | *View on GitHub*

Trained a neural network to identify traffic signs in photographs with 95% accuracy. Used OpenCV to preprocess and resize images from the GTSRB dataset, then built a CNN in TensorFlow with multiple convolutional and pooling layers. I had to experiment with different configurations like filter sizes, dropout rates, and hidden layer sizes to find what worked. The difficult part was avoiding overfitting, but adding dropout layers helped significantly.

PageRank

Tools: Python · Markov Chains · Iterative Algorithms | *View on GitHub*

Built a PageRank algorithm to rank web pages by importance based on link structure and the damping factor. I implemented both a sampling method that simulates a random surfer following links across pages and an iterative formula approach that converges to calculate accurate rankings. The sampling method estimates PageRank by tracking how often a random surfer visits each page, while the iterative approach uses the mathematical formula to refine rankings until they stabilize. Both methods produced similar results, which helped me understand how different computational approaches can solve the same problem.

Workout Tracker Database

Tools: SQL · SQLite · Schema Design · Views · Indexes | *View on GitHub*

Designed a SQLite database schema to manage user workout information, tracking users, exercises, workouts, and progress. Created four main tables with relationships and constraints: users store basic identifiers, workouts record sessions with date and duration, exercises are categorized by muscle group and difficulty level, and progress logs calories burned per user per workout. Additionally, I added indexes on frequently queried columns and built three views to simplify common queries like retrieving a user's total calories or detailed workout information. The main challenge was determining the right relationships between entities since the schema assumes individual workouts, meaning collaborative sessions would need a many-to-many redesign.

Client Websites *with Superb Digital*

A selection of WordPress sites built and delivered for clients through Superb Digital. Each taken from a brief through to a live, client-approved result.

Gents Pride — *gentspride.co.za*

Built a WordPress website for a barbershop in Bonteheuwel, Cape Town with appointment booking, service listings, brand partnerships, and social media integration. The site showcases their 25 year history, barber training programs, and community competitions while allowing customers to book cuts online. Used Elementor for the design and integrated Setmore for appointment scheduling to streamline their booking process.

Township Luxury — *townshipluxury.co.za*

Built the full e-commerce store for Township Luxury, a fashion/streetwear brand, as a side project for a friend during my first month at Temo Digital.

Opian Capital — *opiancapital.com*

Built a website for a private equity and wealth management firm specializing in tech-driven investments and high net worth advisory. Designed the site using Elementor with secure contact forms for consultation requests and integrated analytics to track client inquiries.

MetroWorld Kids — *metroworldkidscpt.co.za*

Built a website for a non-profit youth organization working across 30 schools in Cape Town. Designed the site using Elementor with sections for volunteering, child sponsorships, partnerships, and donations. Integrated donation functionality and volunteer sign-up forms to streamline community engagement.

CERTIFICATIONS

All certifications completed with 73%–100% pass marks from Harvard, MIT, Georgia Tech, NYU, Microsoft, and AWS.

Certification	Institution	Mark	Date	Links	Description
Microsoft Azure Fundamentals	Microsoft	**73%**	Nov 2025	View Certificate	Core Azure services, cloud concepts, security, compliance, and pricing models.
AWS Certified AI Practitioner	Amazon Web Services	**81%**	Oct 2025	View Certificate	AI and ML on AWS, Generative AI, EC2, Lambda, IAM, CloudFormation, S3, Responsible AI.
Prompt Engineering for Microsoft 365 Copilot	Microsoft	**100%**	Jun 2025	View Certificate	Cutting-edge LLM integration, enterprise-grade prompt design, real-world deployment.
CS50 Cybersecurity	Harvard University	**81%**	Nov 2024	View Certificate	Hands-on labs, secure coding, and systems-level security concepts.
CS50 SQL	Harvard University	**95%**	Nov 2024	View Certificate	Relational databases, joins, indexing, and query optimisation.
Cybersecurity Specialization	NYU Tandon	**87%**	Feb 2024	View Certificate	Multi-course specialisation: threat modelling, cryptography, and network security.
CS50AI	Harvard University	**86%**	Feb 2024	View Certificate	Advanced AI concepts, Python, search algorithms, neural networks.
Data Science: R Basics	Harvard University	**94%**	Mar 2024	View Certificate	Introductory R programming and data wrangling.
Microservices & Deployment with ASP.NET	Board Infinity	**95%**	Dec 2023	View Certificate	Service architecture, containerisation, and CI/CD pipelines.
Java II	Georgia Tech	**83%**	Aug 2023	View Certificate	Object-oriented design, data structures, and algorithmic thinking.
Java I	Georgia Tech	**98%**	Jun 2023	View Certificate	Syntax, basic OOP principles, and Java tooling.
Computational Thinking & Data Science	MIT	**94%**	May 2023	View Certificate	Python-based modelling, simulations, and data analysis.
Intro to CS & Python	MIT	**88%**	Mar 2023	View Certificate	Foundational programming, control flow, and problem-solving.
Web Development	Stellenberg High School	**90%**	Mar 2022	View Certificate	HTML, CSS, and basic web design principles.